

INFO5990 26s1 Final Practice

USYD各科往届题库



W2 Aligning IT Investments with Organisational Value

Q1. Short Answer Semester 2 2020 Final (modified)

A modified scenario asks you to advise a mobility start-up whose app, transport-card integration and maintenance processes no longer support its sustainability goal. Identify two relevant concepts from this week (business goals, objectives, strategy, IT operating model, enterprise architecture, cloud service models, IT value) and explain how each should shape the decision.

Q2. MCQ

Which statement best distinguishes a business goal from a business objective?

- (A) A goal is broad direction; an objective is specific and measurable
- (B) A goal is always technical; an objective is always financial
- (C) A goal is written after implementation; an objective is written after audit
- (D) They are identical terms

Q3. MCQ

Which TOGAF domain primarily describes core processes, governance and organisational strategy?

- (A) Technology architecture
- (B) Application architecture
- (C) Business architecture
- (D) Physical security architecture

Q4. True/False

True or false: A business objective can be judged without any measurable target if the IT strategy sounds persuasive. Briefly justify your answer.

Q5. Short Answer

Distinguish a business goal, a business objective and an IT strategy for the mobility start-up. Give one example of each.

Q6. Table/Diagram

Build a four-row table mapping one weakness in the mobility start-up to an EA or IT operating-model response and a KPI.

Q7. Calculation

The start-up scores three options using $\text{Value} = \text{Benefit} - \text{Cost} - \text{Risk}$. Option A app rewrite: 90-45-20. Option B transport-card API integration: 75-25-15. Option C marketing campaign: 60-20-30. Calculate each value and choose the best first investment.

Q8. Short Answer

Write a two-sentence executive recommendation to the CEO that links the transport-card problem to business value and EA.

Q9. Long Question

Create a concise benefits-realisation plan for the mobility start-up: baseline, target, owner, initiative and review cadence.

Q10. Long Question

Challenge: A director argues that the card API is a technical detail and should not be discussed by executives. Respond using IT value alignment.



W3 IT Lifecycle and Project Management Essentials

Q1. Short Answer Semester 1 2024 Final (modified)

A modified scenario asks you to advise a healthcare software provider moving from delayed Waterfall releases to DevSecOps. Identify two relevant concepts from this week (Waterfall, Agile, DevOps, DevSecOps, ITSM, VBA, risk responses, triple constraint) and explain how each should shape the decision.

Q2. MCQ

When requirements are uncertain and frequent user feedback is needed, which delivery model is usually most suitable?

- (A) Pure Waterfall
- (B) Iterative Agile delivery
- (C) Big-bang replacement with no pilots
- (D) Manual operations only

Q3. MCQ

Which item is not one of the classic project triple constraints?

- (A) Scope
- (B) Time
- (C) Cost
- (D) Quality

Q4. True/False

True or false: Moving from Waterfall to Agile automatically creates DevSecOps maturity if developers hold daily stand-ups. Briefly justify your answer.

Q5. Short Answer

Compare Waterfall and Agile for the healthcare provider in terms of feedback timing, risk visibility and delivery cadence.

Q6. Table/Diagram

Make a table with four lifecycle practices: backlog refinement, CI, change control and incident management. State the risk each controls.

Q7. Calculation

A module is estimated at 240 person-hours. Three developers can each contribute 30 productive hours/week. Hourly loaded cost is \$95 and a 15% contingency is required. Calculate duration and budget.

Q8. Short Answer

Draft two acceptance criteria for a secure patient-record release moving from Waterfall to Agile.

Q9. Long Question

Recommend a hybrid lifecycle governance approach that supports Agile delivery while preserving healthcare compliance.

Q10. Long Question

Challenge: Management wants to skip ITSM because DevOps teams will own everything. Explain the risk.



W4 Professional Communication and Stakeholder Management

Q1. Short Answer Semester 2 2023 Final (modified)

A modified scenario asks you to advise a distributed hospital records project with clinicians who have high authority and technical interest. Identify two relevant concepts from this week (stakeholder mapping, power-interest matrix, clarity, brevity, objectivity, conflict resolution, distributed teamwork) and explain how each should shape the decision.

Q2. MCQ

In a power-interest matrix, a stakeholder with high power but low interest should usually be:

- (A) Ignored until the end
- (B) Kept satisfied with concise decision-focused updates
- (C) Given every technical log file
- (D) Removed from governance

Q3. MCQ

A report full of repeated benefits and vague claims mainly fails which writing qualities?

- (A) Brevity and precision
- (B) Encryption and hashing
- (C) Load and stress
- (D) Slicing and dicing

Q4. True/False

True or false: A high-power, low-interest clinician should receive long technical design documents every day to maximise transparency. Briefly justify your answer.

Q5. Short Answer

Distinguish clarity, precision, brevity and objectivity in a stakeholder update about the hospital records system.

Q6. Table/Diagram

Create a stakeholder plan for four groups: senior clinicians, nurses, IT security, and patients/advocates. Include power/interest and communication method.

Q7. Calculation

Score stakeholder priority as Power + Interest + Impact, each 1-5. Senior clinician 5,2,5; ward nurse 3,5,5; security lead 4,4,4; vendor 3,3,3. Rank the top two.

Q8. Short Answer

Write a three-line stakeholder update: status, risk, decision needed.

Q9. Long Question

Design a communication plan for a globally distributed team with low technical-literacy decision makers.

Q10. Long Question

Challenge: A senior clinician resists attending demos. Use the power-interest matrix to justify your engagement approach.



W5 Evidence-Based Practice and Estimation

Q1. Short Answer Semester 2 2024 Final (modified)

A modified scenario asks you to advise a research team validating a survey and estimating an ERP implementation. Identify and explain the relevant concepts from this week (reliability, validity, evidence quality, KPIs, estimation methods, function points, estimation) and explain how each should shape the decision.

Q2. MCQ

A survey that gives consistent results but measures the wrong construct is:

- (A) Valid but not reliable
- (B) Reliable but not valid
- (C) Neither repeatable nor measurable
- (D) A disaster recovery plan

Q3. MCQ

Which estimation approach decomposes the project into deliverables and adds estimates for each part?

- (A) Sum of the parts
- (B) Random allocation
- (C) Pivot analysis
- (D) Social engineering

Q4. True/False

True or false: A survey can be reliable but not valid. Briefly justify your answer.

Q5. Short Answer

Distinguish reliability and validity for an employee-satisfaction survey and give one threat to each.

Q6. Table/Diagram

Create a four-row evidence-quality checklist for deciding whether an estimate is credible.

Q7. Calculation

An ERP estimate has components: configuration 160h, migration 90h, testing 70h, training 40h. Rate is \$110/h; licenses are \$18,000; contingency is 12% of labour+licenses. Calculate total budget.

Q8. Short Answer

Rewrite this weak claim into an evidence-based claim: 'The new dashboard will definitely improve decisions a lot.'

Q9. Long Question

Explain how you would triangulate evidence before recommending the ERP budget.

Q10. Long Question

Challenge: A sponsor asks you to remove uncertainty ranges to make the business case look stronger. Respond professionally.



W6 IT Governance

Q1. Short Answer Semester 2 2020 Final (modified)

A modified scenario asks you to advise a multinational firm needing governance over cloud migration, service security controls. Identify two relevant concepts from this week (COBIT, ITIL, ISO 27001, NIST CSF, audit, control models, governance decision rights) and explain how each should shape the decision.

Q2. MCQ

Which framework is most directly associated with IT service management practices such as incident, problem and change management?

- (A) ITIL
- (B) COBIT
- (C) ISO 27001
- (D) NIST CSF

Q3. MCQ

Which governance framework focuses strongly on aligning IT goals, processes, controls and performance metrics?

- (A) COBIT
- (B) ITIL
- (C) TOGAF
- (D) ISO 27001

Q4. True/False

True or false: IT governance is the same thing as IT operations because both involve technology decisions. Briefly justify your answer.

Q5. Short Answer

Distinguish COBIT, ITIL, ISO 27001 and TOGAF for a cloud migration.

Q6. Table/Diagram

Make a governance table with four decisions: cloud vendor, data access exception, service outage priority, architecture standard exception. Assign a likely forum/owner.

Q7. Calculation

Risk priority = likelihood x impact. Cloud misconfiguration 4x5, weak SLA 3x4, duplicate data 4x3, unclear ownership 5x3. Rank the top two governance risks.

Q8. Short Answer

Write one audit finding with condition, criterion, risk and corrective action.

Q9. Long Question

Recommend a governance approach combining COBIT and ITIL for a multinational service platform.

Q10. Long Question

Challenge: A team wants to bypass governance because it slows delivery. Give a balanced response.



W7 Organisational Change Management

Q1. Short Answer Semester 1 2020 Final (modified)

A modified scenario asks you to advise an ERP rollout where staff keep using spreadsheets and estimates remain low. Identify two relevant concepts from this week (ADKAR, McKinsey 7S, Kotter, adoption metrics, resistance, leadership) and explain how each should shape the decision.

Q2. MCQ

In ADKAR, employees who understand the system but still refuse to use it most clearly lack:

- (A) Desire or reinforcement
- (B) Data extraction
- (C) White-box testing
- (D) Pivot operations

Q3. MCQ

McKinsey 7S is useful because it reminds managers to align:

- (A) Only software and servers
- (B) Hard and soft organisational elements
- (C) Only budget and invoices
- (D) Only encryption and passwords

Q4. True/False

True or false: If staff have completed training, the Knowledge step of ADKAR is always sufficient for successful adoption. Briefly justify your answer.

Q5. Short Answer

Distinguish project management and change management in the ERP spreadsheet scenario.

Q6. Table/Diagram

Map four ADKAR elements to one intervention each for the ERP rollout.

Q7. Calculation

Adoption target is 85%. Four departments report 70%, 82%, 91%, 77% active ERP use. Calculate the average and identify departments below target.

Q8. Short Answer

Write a readiness checklist with four items before turning off the old spreadsheet process.

Q9. Long Question

Use McKinsey 7S to diagnose why the ERP adoption is failing.

Q10. Long Question

Challenge: Leadership wants to punish spreadsheet users immediately. Recommend a better reinforcement strategy.



W8 Quality Assurance and Software Testing

Q1. Short Answer Semester 2 2024 Final (modified)

A modified scenario asks you to advise an e-commerce platform preparing for go-live under heavy and sustained load. Identify two relevant concepts from this week (QA, QC, black-box, white-box, UAT, regression, load, stress, soak, defect lifecycle) and explain how each should shape the decision.

Q2. MCQ

Which test checks externally visible behaviour without knowing internal implementation?

- (A) Black-box testing
- (B) White-box testing
- (C) Code path coverage
- (D) Static branch analysis

Q3. MCQ

Which test best checks whether the system remains stable over an extended operating period?

- (A) Soak testing
- (B) Unit testing
- (C) One-off smoke testing
- (D) Pivot testing

Q4. True/False

True or false: Stress testing and load testing are identical because both involve many users. Briefly justify your answer.

Q5. Short Answer

Distinguish QA, QC and testing for an e-commerce release.

Q6. Table/Diagram

Choose four tests for go-live readiness: load, stress, soak, UAT, regression or security. State what each proves.

Q7. Calculation

Expected peak is 1,200 concurrent users. The system passes at 1,500, degrades at 1,900 and fails at 2,200. Calculate headroom above expected at pass level and classify 2,200 testing.

Q8. Short Answer

Write one high-quality bug report for a checkout timeout found during load testing.

Q9. Long Question

Design a minimal test plan for the e-commerce go-live.

Q10. Long Question

Challenge: The product owner wants to skip UAT because all automated tests pass. Explain why that is risky.



W9 Security Management 1

Q1. Short Answer Semester 2 2024 Final (modified)

A modified scenario asks you to advise a company experiencing convincing phishing attempts and suspicious privilege escalation activity. Identify two relevant concepts from this week (information security, cybersecurity, privacy, CIA, ST engineering, APT, incident response) and explain how each should shape the decision.

Q2. MCQ

Which STRIDE category describes pretending to be another user or system?

- (A) Spoofing
- (B) Tampering
- (C) Repudiation
- (D) Denial of service

Q3. MCQ

Which control most directly reduces unauthorised login risk from stolen passwords?

- (A) MFA
- (B) More colourful dashboards
- (C) Ignoring logs
- (D) Removing backups

Q4. True/False

True or false: MFA eliminates the need for user education against social engineering. Briefly justify your answer.

Q5. Short Answer

Distinguish information security, cybersecurity and data privacy in the phishing scenario.

Q6. Table/Diagram

Map four STRIDE threats to one control each for privileged account activity.

Q7. Calculation

Three threats have likelihood/impact: phishing 5/4, privilege escalation 3/5, log deletion 2/4. Compute risk scores and choose the first control focus.

Q8. Short Answer

Write a one-paragraph incident triage note for suspicious privileged login activity.

Q9. Long Question

Develop a threat model summary for the phishing scenario.

Q10. Long Question

Challenge: A manager says phishing is a people problem, not an IT control problem. Respond.



W10 Security Management 2

Q1. Short Answer Semester 2 2020 Final (modified)

A modified scenario asks you to advise a financial services firm migrating sensitive client data to a hybrid cloud. Identify relevant concepts from this week (BCP, DR, RTO, RPO, asset protection, privacy principles, hybrid cloud security strategy) and explain how each should shape the decision.

Q2. MCQ

Which statement best describes disaster recovery?

- (A) A subset of continuity planning focused on restoring IT systems
- (B) A preventive control that eliminates all outages
- (C) A business objective for customer acquisition
- (D) A software testing method for peak traffic

Q3. MCQ

RPO mainly answers which question?

- (A) How much data loss is tolerable?
- (B) How quickly service must be restored
- (C) Which controls prevent all incidents
- (D) Which stakeholder approves the architecture board

Q4. True/False

True or false: BCP and DR are interchangeable terms for restoring servers after a disruption. Briefly justify your answer.

Q5. Short Answer

Distinguish RTO and RPO for banking-data migration.

Q6. Table/Diagram

Create a four-row hybrid-cloud control table: identity, data, network, recovery.

Q7. Calculation

A service has RTO 4 hours and RPO 30 minutes. An incident restores service after 5.5 hours with 20 minutes of lost transactions. Which target failed?

Q8. Short Answer

Write two privacy requirements for migrating sensitive banking data to hybrid cloud.

Q9. Long Question

Recommend a BCP/DR approach for the banking cloud migration.

Q10. Long Question

Challenge: A vendor promises 99.9% uptime but refuses restore-test evidence. Explain the governance concern.



W11 Ethics and Professional Responsibility in IT

Q1. Short Answer Semester 2 2024 Final (modified)

A modified scenario asks you to advise a smart-city AI initiative raising privacy, bias and unequal benefit concerns. Identify relevant concepts from this week (ACS Code of Ethics, public interest, societal ethics, AI fairness, privacy, breach of trust, accountability) and explain how each should shape the decision.

Q2. MCQ

Which ethical concern is most directly raised by an AI recruitment system trained on biased historical hiring data?

- (A) Automated discrimination
- (B) Reduced server utilisation
- (C) Shorter release cycles
- (D) More consistent invoice numbering

Q3. MCQ

When management asks staff to hide a serious data breach, the strongest professional response is to:

- (A) Follow the request silently
- (B) Escalate through proper channels and protect affected users
- (C) Delete logs
- (D) Post rumours online

Q4. True/False

True or false: If an AI system is accurate on average, fairness concerns can be ignored. Briefly justify your answer.

Q5. Short Answer

Distinguish individual professional ethics and societal ethics in the smart-city AI scenario.

Q6. Table/Diagram

Map privacy, algorithmic bias, and equitable benefit to one ethical principle and one mitigation.

Q7. Calculation

A bias audit finds false-positive rates of 4% in Group A and 11% in Group B. Calculate the gap and state one governance action.

Q8. Short Answer

Write a short ethical recommendation for deploying smart-city sensors.

Q9. Long Question

Use an ethical decision process to assess the smart-city initiative.

Q10. Long Question

Challenge: The department argues efficiency benefits justify collecting all possible citizen data. Respond ethically.



W12 Data-Driven Decision Making and Business Intelligence

Q1. Short Answer Semester 2 2024 Final (modified)

A modified scenario asks you to advise a retail chain using BI to reduce stockouts and rebalance inventory across stores. Advise the retail chain using two relevant concepts from this week (data, information, knowledge, ETL, data warehouse, BI, OLAP, dashboard, data quality, analytics types) and explain how each should shape the decision.

Q2. MCQ

In OLAP, which operation rotates or reorients dimensions for comparison?

- (A) Pivot
- (B) Roll up
- (C) Drill down
- (D) Slice

Q3. MCQ

Which option best describes information rather than raw data?

- (A) Unprocessed POS transaction rows
- (B) Summarised weekly stockout rate by store
- (C) A barcode number with no context
- (D) A random server log fragment

Q4. True/False

True or false: A dashboard becomes business intelligence only when it supports interpretation and action, not merely when it displays data. Briefly justify your answer.

Q5. Short Answer

Distinguish data, information and knowledge using the retail stockout scenario.

Q6. Table/Diagram

Map four OLAP operations to retail inventory questions.

Q7. Calculation

Store A has 18 stockout days out of 120 product-days; Store B has 9 out of 150; Store C has 20 out of 200. Calculate stockout rates and rank worst to best.

Q8. Short Answer

Write two dashboard design improvements for a retail inventory manager.

Q9. Long Question

Explain how a data warehouse and BI tools support inventory decisions.

Q10. Long Question

Challenge: A manager wants to reorder solely from a predictive score. Explain the decision risk.



W13 Integrated Review and Professional Foundations

Q1. Short Answer Semester 1 2020 Final (modified)

A modified scenario asks you to advise a graduate cybersecurity consultant handling confidential client records. Identify relevant concepts from this week (professionalism, ACS ethical duties, non-technical skills, career development, organisational structures) and explain how each should shape the decision.

Q2. MCQ

Which action best demonstrates IT professionalism in a client security incident?

- (A) Wait until all facts are perfect before telling anyone
- (B) Escalate promptly with evidence, protect confidentiality, and avoid speculation
- (C) Share details with peers to obtain informal opinions
- (D) Focus only on technical remediation and ignore communication

Q3. MCQ

A matrix organisation is most likely to create which reporting pattern?

- (A) No managers at all
- (B) Only regional reporting
- (C) Dual functional and project reporting
- (D) All decisions made by one founder

Q4. True/False

True or false: Professional practice questions can be answered well using only generic definitions. Briefly justify your answer.

Q5. Short Answer

Distinguish professionalism, ethics and a code of conduct for a graduate cybersecurity consultant.

Q6. Table/Diagram

Create a four-row review table linking professionalism, governance, security and communication to one exam scenario risk.

Q7. Calculation

A review question weights professionalism 25%, governance 25%, security 25%, communication 25%. A sample answer scores 3,2,4,1 out of 5. Calculate the weighted score out of 5.

Q8. Short Answer

Write a two-sentence professional escalation note for a suspected data breach.

Q9. Long Question

Write an integrated response to a scenario involving breach handling, stakeholder communication and ethical responsibility.

Q10. Long Question

Challenge: A supervisor asks you to stay silent about a harmful system weakness until after a sales demo. Respond professionally.



Answer Key

W2 Aligning IT Investments with Organisational Value

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. A goal is broad direction; an objective is specific and measurable.
- Q3.** C. Business architecture.
- Q4.** False. An objective should be measurable; persuasive strategy language is not enough to evaluate value realisation.
- Q5.** Goal: reduce transport emissions or improve sustainable mobility. Objective: increase active riders by a measurable percentage/time. IT strategy: improve app reliability, integration and analytics to support that objective.
- Q6.** Expected rows may include poor card integration -> application/data architecture integration -> failed transfer rate; app defects -> service ownership/ITSM -> incident rate; bike loss -> data/control process -> loss rate; weak engagement -> BI/CRM capability -> active-user retention.
- Q7.** A = 25, B = 35, C = 10. Option B ranks first because it has the highest net value and directly fixes a business-IT integration weakness.
- Q8.** Should mention that integration failure undermines the sustainability/customer-adoption objective, and recommend an EA-led application/data integration plan with ownership, API governance and KPI monitoring.
- Q9.** Credit for a baseline (active users/failed integrations), SMART target, named owner, initiative such as API integration/app reliability, and monthly/quarterly review with corrective action.
- Q10.** A strong response explains that the API is a business capability dependency; executives decide priorities, risk appetite and funding when technical integration affects customer adoption and strategic goals.

W3 IT Lifecycle and Project Management Essentials

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** B. Iterative Agile delivery.
- Q3.** D. Quality.
- Q4.** False. Agile ceremonies do not create DevSecOps; secure CI/CD, automation, shared ownership, testing and governance are also needed.
- Q5.** Waterfall has late feedback, late risk discovery and large releases; Agile uses frequent feedback, early risk visibility and incremental delivery.
- Q6.** Backlog refinement controls requirement ambiguity; CI controls integration defects; change control controls production disruption/compliance; incident management controls service interruption and learning.
- Q7.** Weekly capacity = 90 hours; duration = $240/90 = 2.67$ weeks, so about 3 weeks. Labour cost = $240 \times 95 = \$22,800$; with 15% contingency = \$26,220.
- Q8.** Criteria should be testable, e.g. authorised clinicians can retrieve records within a threshold, and all privileged access is logged/MFA-protected before release.
- Q9.** Good answers combine Agile iterations with architecture/security gates, documented requirements for regulated features, automated tests, change control and audit evidence.
- Q10.** DevOps improves delivery/operations collaboration, but ITSM provides service processes, incident/problem/change discipline, SLAs and accountability needed for reliable clinical systems.

W4 Professional Communication and Stakeholder Management

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** B. Kept satisfied with concise decision-focused updates.
- Q3.** A. Brevity and precision.
- Q4.** False. They should be kept satisfied with concise, decision-focused updates and consulted when their authority or clinical workflow is affected.
- Q5.** Clarity makes meaning easy; precision uses exact facts/metrics; brevity removes repetition; objectivity relies on evidence rather than hype or blame.
- Q6.** Senior clinicians often high power/lower interest -> concise decision briefs; nurses high interest -> workshops/demos; security high influence -> design reviews; patients/advocates lower formal power but high impact -> consultation/plain-language



updates.

Q7. Senior clinician = 12, ward nurse = 13, security lead = 12, vendor = 9. Ward nurse ranks first; senior clinician lead tie next and need different engagement styles.

Q8. Must include current progress, a concrete risk such as unclear clinical workflow or access control, and a date with owner/date.

Q9. Credit for channels, cadence, plain-language artefacts, async records, decision logs, escalation path and feedback loops.

Q10. Keep satisfied through short business/clinical impact briefings, involve only in key workflow decisions, and avoid overwhelming them with implementation detail.

W5 Evidence-Based Practice and Estimation

Q1. Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.

Q2. B. Reliable but not valid.

Q3. A. Sum of the parts.

Q4. True. It may produce consistent results while measuring the wrong construct.

Q5. Reliability concerns consistency; validity concerns measuring satisfaction accurately. Threats include ambiguous questions for reliability and poor construct coverage/social desirability for validity.

Q6. Rows may cover source credibility, sample/relevance, method transparency, historical comparison, assumptions and uncertainty range.

Q7. Hours = 360; labour = \$39,600; subtotal = \$57,600; contingency = \$6,912; total = \$64,512.

Q8. Should add evidence and measurable outcome, e.g. pilot data suggests managers identify stockouts faster; target is 20% reduction in stockout response time, with assumptions and limitations.

Q9. Use historical project data, expert judgement, vendor quote, WBS estimates, prototype findings and risk reserve; compare and document assumptions.

Q10. State that hiding uncertainty weakens decision quality; present ranges, assumptions and mitigation while still giving a recommendation.

W6 IT Governance

Q1. Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.

Q2. A. ITIL.

Q3. A. COBIT.

Q4. False. Governance sets decision rights, accountability and monitoring; operations execute and manage services.

Q5. COBIT aligns IT and governance controls; ITIL manages services; ISO 27001 structures security management; TOGAF structures architecture domains.

Q6. Examples: steering committee/vendor risk; data owner/security/privacy for access; ITSM incident manager for outage; architecture board for standards.

Q7. Misconfiguration = 20, unclear ownership = 15, weak SLA = 12, duplicate data = 12. Top two: misconfiguration and ownership.

Q8. Must state what was observed, which policy or standard it violates, the business or security risk, and a concrete remediation owner and due date.

Q9. COBIT for decision rights, control objectives and performance metrics; ITIL for incident/problem/change/service catalogue; link both to business objectives and KPIs.

Q10. Acknowledge speed concerns, propose proportionate lightweight gates/automation, and explain governance prevents unmanaged risk, duplication and compliance failure.

W7 Organisational Change Management

Q1. Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.

Q2. A. Desire or reinforcement.

Q3. B. Hard and soft organisational elements.

Q4. False. Adoption also requires awareness, desire, ability in real work and reinforcement.



- Q5.** Project management delivers the ERP within scope/time/cost; change management ensures staff adopt it using spreadsheets.
- Q6.** Awareness: explain why; Desire: sponsor/user benefits; Knowledge: role-based training; Ability: hands-on enforcement; KPI/incentive/support review.
- Q7.** Average = 80%. Departments at 70%, 82% and 77% are below target; targeted ability/reinforcement interventions are needed.
- Q8.** Include data migration checked, role permissions, process ownership, training/support, help desk readiness and fall-back/issue escalation.
- Q9.** Credit for aligning systems with skills, staff, style, shared values, structure and strategy; identify mismatches such as incentives or workflow fit.
- Q10.** First diagnose barriers, support ability, align incentives and make ERP the easier trusted path; sanctions alone may increase resistance.

W8 Quality Assurance and Software Testing

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. Black-box testing.
- Q3.** A. Soak testing.
- Q4.** False. Load testing checks expected workload; stress testing pushes beyond limits to find breaking points.
- Q5.** QA improves the process to prevent defects; QC inspects outputs; testing executes checks to find defects and provide evidence.
- Q6.** Load expected traffic; stress breaking point; soak long-term stability; UAT business fit; regression old features; security vulnerabilities.
- Q7.** Pass headroom = $(1500-1200)/1200 = 25\%$. 2,200 is stress testing because it seeks the failure/breaking point.
- Q8.** Should include environment, steps, expected/actual result, evidence, severity/priority and business impact.
- Q9.** Include scope, risks, environment/data, test types, entry/exit criteria, defect workflow and roles.
- Q10.** Automated tests verify specified behaviours; UAT checks real business workflow, usability and acceptance from users.

W9 Security Management 1

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. Spoofing.
- Q3.** A. MFA.
- Q4.** False. MFA reduces credential misuse but social engineering can still target approvals, data disclosure and process bypass.
- Q5.** Information security protects information assets; cybersecurity protects digital systems/networks; privacy governs appropriate personal-data handling.
- Q6.** Spoofing -> MFA; tampering -> change control/integrity logs; repudiation -> audit logs; information disclosure -> access control/encryption; DoS -> rate limiting; elevation -> PAM.
- Q7.** Phishing = 20, privilege escalation = 15, log deletion = 8. First focus: phishing controls such as MFA, awareness, reporting and email filtering.
- Q8.** Include time/source/account, evidence, immediate containment, preservation of logs, stakeholder notification and next investigation step.
- Q9.** Identify assets, actors, entry points, trust boundaries, STRIDE threats, controls and residual risk.
- Q10.** Explain socio-technical risk: training matters, but technical controls, monitoring, process and culture are also required.

W10 Security Management 2

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. A subset of continuity planning focused on restoring IT systems.
- Q3.** A. How much data loss is tolerable?.
- Q4.** False. DR focuses on IT restoration; BCP covers continued business operations and dependencies.
- Q5.** RTO is maximum tolerable restoration time; RPO is maximum tolerable data loss.



- Q6.** Identity: MFA/PAM/federation; data: encryption/classification/DLP; network: segmentation/private links; backup/DR runbooks.
- Q7.** RTO failed because 5.5 hours exceeds 4 hours. RPO passed because 20 minutes is less than 30 minutes.
- Q8.** Examples: lawful basis/consent and transparency; data minimisation; encryption and access controls; breach data residency; retention/deletion.
- Q9.** Include BIA, critical processes, RTO/RPO, backup strategy, failover, tested runbooks, roles, comms and supplier SLAs.
- Q10.** SLA promises need assurance; without test evidence the organisation cannot verify recoverability, residual risk or compliance.

W11 Ethics and Professional Responsibility in IT

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. Automated discrimination.
- Q3.** B. Escalate through proper channels and protect affected users.
- Q4.** False. Aggregate accuracy can hide biased harms to subgroups and unfair distribution of benefits.
- Q5.** Individual ethics concerns the practitioner's duties and integrity; societal ethics concerns public impact, equity, rights and collective welfare.
- Q6.** Privacy -> rights/public interest -> minimisation/consent/security; bias -> fairness -> representative data/audit; equity -> public good/justice -> inclusive consultation and access measures.
- Q7.** Gap = 7 percentage points. Action: investigate data/model causes, suspend high-risk use or add human review, monitor by subgroup and document accountability.
- Q8.** Should name stakeholders, principle, risk and action such as privacy-by-design, bias monitoring, transparency and appeal/oversight.
- Q9.** Identify stakeholders, harms/benefits, duties/rights, options, ACS/public interest principle, recommendation and monitoring.
- Q10.** Efficiency does not override proportionality, minimisation, privacy, fairness and public trust; collect only justified data with safeguards.

W12 Data-Driven Decision Making and Business Intelligence

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** A. Pivot.
- Q3.** B. Summarised weekly stockout rate by store.
- Q4.** True. BI should connect data to decision-making through context, KPIs, analysis and action.
- Q5.** Data: raw POS/inventory records; information: summarised stockout rate; knowledge: interpretation that supplier lead time or allocation causes stockouts and what action to take.
- Q6.** Slice by store/category; dice store+week+category; drill down from region to store/SKU; roll up SKU to category; pivot store vs product dimensions.
- Q7.** A = 15%, B = 6%, C = 10%. Worst to best: A, C, B.
- Q8.** Examples: show threshold alerts with owner/action, trend and drill-down by SKU/store, forecast error, lead time and exception list.
- Q9.** Warehouse integrates POS/inventory/supplier history; BI provides consistent KPIs, trend/exception analysis and drill-down to operational action.
- Q10.** Scores require context, data-quality checks, uncertainty, supplier constraints and ethical/business review; decisions need monitoring and human accountability.

W13 Integrated Review and Professional Foundations

- Q1.** Use two scenario-specific concepts, define each accurately, and connect each to an action or risk. High-scoring answers name the stakeholder impact, governance or ethical duty, and a practical mitigation rather than giving generic definitions.
- Q2.** B. Escalate promptly with evidence, protect confidentiality, and avoid speculation.
- Q3.** C. Dual functional and project reporting.
- Q4.** False. Strong INFO5990 answers combine definitions with scenario evidence, stakeholder impact and justified action.



Q5. Professionalism is competent accountable conduct; ethics guides right action under value conflict; a code of expected behavioural standards.

Q6. Rows should connect each theme to a scenario-specific risk and mitigation, such as confidentiality, escalation stakeholder updates.

Q7. Weighted score = $(3+2+4+1)/4 = 2.5$ out of 5. Biggest improvement area is communication.

Q8. Should state evidence without speculation, immediate containment/escalation, confidentiality and concern for affected users/legal duties.

Q9. Credit for defining the issue, protecting users, escalating with evidence, applying ACS/public interest, proposing controls and communication.

Q10. Explain duty of honesty/public interest, document risk, escalate through proper channels and propose a proportionate disclosure/remediation plan.